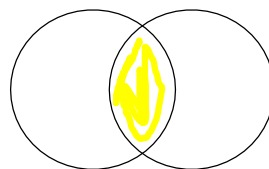


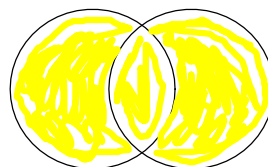
OPERAZIONI INSIEMISTICHE

$\forall A, B, E$ insiemi : $A, B \subseteq E$:

• **INTERSEZIONE** : $A \cap B = \{x | (x \in A) \wedge (x \in B)\}$
 $\downarrow$
 "E"

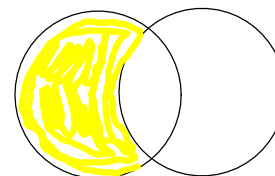


• **UNIONE** : $A \cup B = \{y | (y \in A) \vee (y \in B)\}$

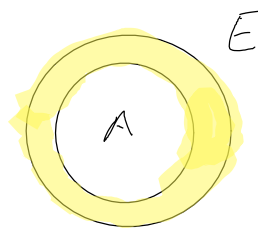


• **DIFFERENZA** : $A \setminus B = \{z | (z \in A) \wedge (z \notin B)\}$
 $\downarrow$
 MENO

$$A \setminus B \neq B \setminus A$$



• **COMPLEMENTARE** : $C(A) = E \setminus A$
 $\downarrow$
 E
 (complementare di A in E)



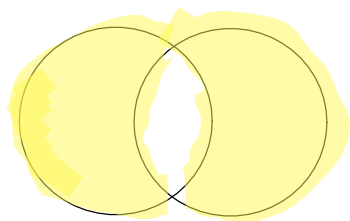
$$\cap (A \setminus B) \cup (B \setminus A)$$

• **DIFFERENZA SIMMETRICA:**

↓

serve nelle ALGEBRE di BOOL!

$$A \oplus B = \left\{ (A \cup B) \setminus (A \cap B) \right\}$$



TUTTE queste operazioni $\in \mathcal{P}(E)$.

• **PRODOTTO CARTESIANO** = $\{ (a, b) \mid (a \in A) \wedge (b \in B) \}$

↓
 E

